

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **25**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 25**Solve for y .

$$-\frac{5}{2}y + \frac{5}{2} = -\frac{5}{3}y - 6$$

Simplify your answer as much as possible.

$$y = \boxed{}$$

Question 2 of 25Solve for x .

$$\frac{8x - 9}{2} = 41 - \frac{7x - 6}{6}$$

Simplify your answer as much as possible.

$$x = \boxed{}$$

Question 3 of 25Solve for W .

$$5 = LW$$

$$W = \boxed{}$$

Question 4 of 25

A washer and a dryer cost \$787 combined. The washer costs \$37 more than the dryer. What is the cost of the dryer?

Question 5 of 25

Two trains leave the station at the same time, one heading west and the other east. The westbound train travels at 85 miles per hour. The eastbound train travels at 75 miles per hour. How long will it take for the two trains to be 288 miles apart?

Do not do any rounding.

hours

hours

Question 6 of 25

An item is regularly priced at \$69. It is now priced at a discount of 70% off the regular price.

Use a calculator to find the price now.

Question 7 of 25

Write in terms of i .

Simplify your answer as much as possible.

$$\sqrt{-36}$$

Question 8 of 25

Add.

$$(1 - 5i) + (2 - i)$$

Write your answer as a complex number in standard form.

Question 9 of 25

Multiply.

$$(-4 + 5i)(-2 + 2i)$$

Write your answer as a complex number in standard form.

Question 10 of 25

Divide.

$$\frac{1 - 2i}{-2 - 6i}$$

Write your answer as a complex number in standard form.

Question 11 of 25

Simplify the complex number i^{37} as much as possible.

Question 12 of 25

Solve for y .

$$y^2 - y - 12 = 0$$

$$y =$$

Question 13 of 25

Solve for y .

$$5y^2 = -16y - 3$$

$$y =$$

Question 14 of 25

Solve the quadratic equation by completing the square.

$$x^2 + 2x - 5 = 0$$

(a) Choose the appropriate form and fill in the blanks with the correct numbers.

☐ $(x + \underline{\hspace{1cm}})^2 = \underline{\hspace{1cm}}$

☐ $(x - \underline{\hspace{1cm}})^2 = \underline{\hspace{1cm}}$

(b) Solve the equation. Simplify your answer as much as possible. If there is more than one solution, separate them with commas.

$x = \underline{\hspace{3cm}}$

Question 15 of 25

Use the quadratic formula to solve for x .

$$4x^2 - 9x - 1 = 0$$

(If there is more than one solution, separate them with commas.)

$x =$

Question 16 of 25

Solve for u .

$$|4u + 4| - 8 = -36$$

Question 17 of 25

Solve for w , where w is a real number.

$$\sqrt{2w - 3} + 1 = 4$$

Question 18 of 25

Solve for w , where w is a real number.

$$\sqrt[4]{w} = 2$$

$w =$

Question 19 of 25

Write inequalities to represent the situations below.

- (a) The cruising speed of the bullet train will be no less than 160 miles per hour.

Write an inequality using s to represent the train's cruising speed (in miles per hour).

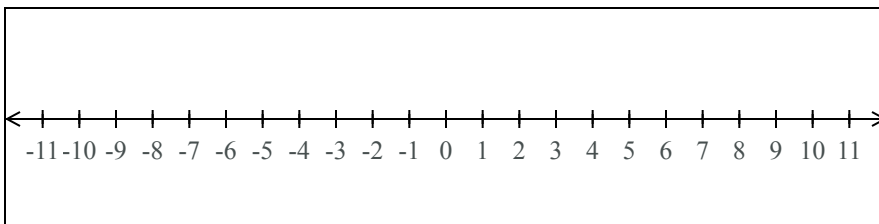
- (b) To be eligible for the senior admission rate, a visitor must be at least 55 years old.

Write an inequality using y to represent the age (in years) of a visitor who gets the senior rate.

Question 20 of 25

Graph the inequality below on the number line.

$$a < -6$$



Question 21 of 25

Write the following as an inequality.

x is greater than -5 and

less than or equal to 4

Use x only once in your inequality.

Question 22 of 25

Solve the inequality for u .

$$21 - 5u > 6$$

Simplify your answer as much as possible.

Question 23 of 25

Solve the compound inequality.

$$4x - 6 \geq 6 \quad \text{or} \quad 3x - 4 \geq 14$$

Write the solution in interval notation.

If there is no solution, enter $\emptyset$.

Question 24 of 25

Solve.

$$5|x - 7| + 9 < 49$$

Question 25 of 25

Solve for v , where v is a real number.

$$\frac{1}{v^4} = -4$$

If there is more than one solution, separate them with commas.

If there is no solution, click on "No solution".

$v =$

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Question 1 of 25

$$y = \frac{51}{5}$$

Question 2 of 25

$$x = 9$$

Question 3 of 25

$$W = \frac{5}{L}$$

Question 4 of 25

\$375

Question 5 of 25

1.8 hours

Question 6 of 25

\$20.70

Question 7 of 25

$$6i$$

Question 8 of 25

$$3 - 6i$$

Question 9 of 25

$$-2 - 18i$$

Question 10 of 25

$$\frac{1}{4} + \frac{1}{4}i$$

Question 11 of 25

$$i$$

Question 12 of 25

$$y = 4, -3$$

Question 13 of 25

$$y = -\frac{1}{5}, -3$$

Question 14 of 25

(a) ☒ $(x + 1)^2 = 6$

☐ $(x - \boxed{})^2 = \boxed{}$

(b) $x = -1 + \sqrt{6}, -1 - \sqrt{6}$

Question 15 of 25

$$x = \frac{9 + \sqrt{97}}{8}, \frac{9 - \sqrt{97}}{8}$$

Question 16 of 25

No solution

Question 17 of 25

$$w = 6$$

Question 18 of 25

$$w = 16$$

Question 19 of 25

(a) The cruising speed of the bullet train will be no less than 160 miles per hour.

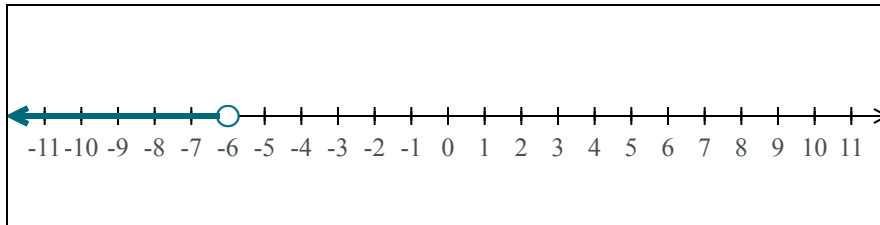
Write an inequality using s to represent the train's cruising speed (in miles per hour).

$$s \geq 160$$

(b) To be eligible for the senior admission rate, a visitor must be at least 55 years old.

Write an inequality using y to represent the age (in years) of a visitor who gets the senior rate.

$$y \geq 55$$

Question 20 of 25**Question 21 of 25**

$$-5 < x \leq 4$$

Question 22 of 25

$$u < 3$$

The answer can also be written as $3 > u$.

Question 23 of 25

$$[3, \infty)$$

Question 24 of 25

$$-1 < x < 15$$

Question 25 of 25

No solution